Sentiment classification of patient medication feedback text

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***Abstract*—Medicine manufacturing company can monitor the effectiveness and side effect of a new medicine by extracting information from patient medication feedback text on social media. A pre-requisite for this process is the patient medication feedback text needs to be segregated into positive sentiment and negative sentiment, as positive sentiment is usually related to effectiveness, whereas negative sentiment is usually related to side effects. However, manually segregating the raw medication feedback text based on sentiment is a labor-intensive task. So, this project proposes solution that automates the sentiment classification of medication feedback text. In this project, five feature representations, namely Valence Aware Dictionary for Sentiment Reasoning (VADER) sentiment lexicon, Bag of Words (BOW), BOW + VADER, Term Frequency – Inverse Document Frequency (TF-IDF) and TF-IDF + VADER, are compared. For VADER alone, no machine learning is involved, whereas for BOW, BOW + VADER, TF-IDF and TF-IDF + VADER, supervised learning methods, such as multinomial naïve Bayes, logistic regression and multi-layer perceptron, are compared. This project found that generally machine learning models outperformed the baseline VADER sentiment lexicon. The best performing model found in this project is multi-layer perceptron with BOW + VADER features, with macro average F1 of 0.8300. Besides building a feasible automated sentiment classification solution, this project also investigates whether combining sentiment lexicon information and BOW or TF-IDF feature representation can improve the performance of the model. From the results of this project, it is found that although adding sentiment lexicon information can be helpful in some cases, but there is no guarantee of improvements in performance because in some cases, adding sentiment lexicon information can add noise to the conventional BOW or TF-IDF feature representation.**

Keywords*—medication feedback, sentiment classification, VADER lexicon, Bag of Words, Term Frequency-Inverse Document Frequency, supervised machine learning

I. Introduction

Medicine manufacturing companies release new medicine to the market after a rigorous and comprehensive research on the effectiveness and side effects of the medicine. However, the new medicine may still have some undiscovered side effect on different group of patients. This is because the research conducted by the medicine manufacturing company may not cover all type of patients, in terms of ethnic, gender, age range, medical history and whether have other background diseases. For the same reason, the medicine is also possible to be less effective on certain group of patients.

So, to know the side effects encountered by patients and the effectiveness of the new medicine after the medicine is released to the market, the medicine manufacturing company can monitor the experience of patients taking the medicine by examining the sentiment of the medicine review written by patients. Those reviews are normally available on social media posts. If the sentiment is positive, then there is less likely that the patient experiences side effect and more likely that the medicine is effective. Oppositely, if the sentiment is negative, then it is likely that the patient experiences side effect or the medicine is less effective. The medicine manufacturing company can then analyse the negative medicine reviews to find out what is the side effect experienced by the patients and analyse the cases where the medicine is less effective.

However, examining the sentiment of each social media post needs a lot of manpower since there might be thousands of social media posts mentioning the medicine. So, some kind of automation is needed to examine the sentiment of the medicine reviews automatically. One way to examine the sentiment of patient medication review automatically is by applying automated sentiment classification on patient medicine review. This brings to the main purpose of this project, which is to build a sentiment classification model to classify the sentiment of a patient medicine review in text, which is the very first step for the medicine manufacturing company to do before they can analyse the side effects and the effectiveness of their new medicine.

II. Problem

The problem of this project is to compare different approaches to build a sentiment classification model to classify the sentiment of a patient medicine review into either positive review or negative review. The patient medicine review text is analysed to determine the sentiment. The approaches for building a sentiment classification model compared in this project are sentiment lexicon method (as baseline), supervised learning method and integrating sentiment lexicon into supervised learning method.

The sentiment lexicon method in this project works by taking the average polarity score of the words in a patient medicine review assigned by Valence Aware Dictionary for Sentiment Reasoning (VADER) sentiment lexicon. If the average polarity score falls in the region of positive polarity, then the sentiment of the review is considered positive, whereas if the average polarity score falls in the region of negative polarity, then the sentiment of the review is considered negative. The sentiment lexicon method does not need any learning and is solely based on VADER sentiment lexicon.

The supervised learning method in this project compares two different feature representations, namely bag of words (BOW) and term frequency-inverse document frequency (TF-IDF). For each feature representation, three machine learning algorithms are compared, namely multinomial naïve Bayes, logistic regression and multi-layer perceptron. The machine learning algorithm learns to classify the sentiment based on a labelled patient medicine review dataset.

This project also tries to integrate sentiment lexicon into supervised learning by generate some features from the VADER sentiment lexicon and feed these features together with BOW or TF-IDF into the machine learning model. These features are the proportion of positive words, the proportion of negative words and the proportion of neutral words. This method incorporates useful lexicon knowledge to the supervised learning model, with the intuition that knowing the lexicon knowledge will help the supervised learning model to classify the sentiment more accurately.

The dataset used in this project is from Kaggle. The owners of the dataset used web scraping to retrieve the patient medicine review from drugs.com and posted the dataset on Kaggle [1]. The attributes in the dataset are drug name, condition, user, date, rating and content. In this project, only rating attribute and content attribute are used. The content attribute is the patient medicine review text, whereas the rating attribute is the rating of patient satisfaction of the medicine, ranging from 1 to 10. In this project, reviews with rating of 1 to 4 is labelled as negative sentiment, whereas reviews with rating of 5 to 10 is labelled as positive sentiment. The labelled patient medicine reviews dataset is the dataset used in this project to build and evaluate the patient medicine review sentiment classifier.

The problem is now turned into a binary classification problem that classifies the patient medication feedback text into positive sentiment or negative sentiment. The distribution of classes is not balanced because the positive sentiment class has 177395 instances and the negative sentiment class has 81787 instances. So, since the class distribution is imbalance, the performance of this problem is evaluated based on macro average F1 of the two classes. Using accuracy or weighted average F1 is not suitable because these two metrics can hide the lower performance on the minority class.

III. Related Works

Allenbrand [2] worked on a task to predict the benefit and side effect of a drug on a three-point scale and five-point scale based on the drug review text written by patients. The research used BOW with unigram and bigram as feature representation and compared three machine learning algorithms, namely naïve Bayes, support vector machine with radial basis function kernel and random forest [2]. The research found that random forest is the best performing classifier, three-point scale works better than five-point scale and unigram works better than bigram [2].

Nair et. al. [3] worked on a sentiment analysis task on user-generated drug reviews from drugs.com, with a three-point scale. The research used word embedding feature representation and trained the classification model using decision tree, support vector machine, random forest, logistic regression and recurrent neural network [3]. The results showed that the model trained with logistic regression is the best performing model in the research [3].

Bottacin et. al. [4] analysed the effectiveness of using sentiment lexicons for sentiment analysis of patient medication feedback text. The sentiment lexicons used in the study are VADER and Emotion English DistilRoBERTa-base (DistilRoBERTa) [4]. The research found that the emotion score assigned using VADER based on the text is strongly correlated to the ratings assigned by the patients [4]. The research also found that the sentiment of the positively and negatively rated feedbacks are associated with different emotions assigned by DistilBERTa [4]. In the research, VADER was used to predict the sentiment of the feedback text and the model had achieved F1 of 0.73 for negative class and F1 of 0.65 for positive class [4].

Azar et. al. [5] compared traditional machine learning algorithms, namely k nearest neighbours, decision tree, random forest and artificial neural network, the ensemble of machine learning algorithms, deep learning algorithm and the ensemble of deep learning algorithms on classifying the sentiment of patient review on medication. In the research, BOW feature representation was used for traditional machine learning algorithms, whereas word embedding was used for deep learning algorithms [5]. In the research, the ten-scale rating was converted to two-scale (below 5 and 5 or above) and three-scale (below 5, 5 and 6, and above 6), and two-scale outperformed three-scale [5]. The research found that the ensemble of deep learning algorithms is the best performing method, and random forest is the best performing traditional machine learning algorithm [5].

Saad et. al. [6] worked on a task to analyse the efficiency of drug based on sentiment analysis of patient drug review on healthcare web forum. In the study, several sentiment lexicons, namely AFINN, TextBlob and VADER, were used to generate the ground truth label, and several machine learning algorithms, namely logistic regression, random forest, extra tree classifier, adaptive boosting classifier and multi-layer perceptron were compared on the task of classifying sentiment of patient drug review using feature representations of term frequency (TF), TF-IDF and the union of TF and TF-IDF [6]. The results showed that logistic regression and multi-layer perceptron performed better than the other machine learning algorithms, with accuracies of more than 0.9.

Table 1 summarizes all aforementioned related works in terms of the best performing model in each literature, the feature representation used and the number of classes in the sentiment classification task.

Table 1 Summary of related works

Literature	Best performing model	Feature representation	Number of classes
[2]	Random forest	BOW	3 (best), 5
[3]	Logistic regression	Word embedding	3
[4]	VADER (lexicon, not machine learning)	Text (not converted to numerical feature representation)	3 (VADER), 5 (DistilBERTa)
[5]	Ensemble of deep learning models (overall), random forest (traditional machine learning)	BOW (traditional machine learning), word embedding (deep learning)	2 (best), 3, 10

[6]	Logistic regression, multi-layer perceptron (equally good)	TF, TF-IDF, union of TF and TF-IDF	3

Since the scope of this project focuses on enhancing traditional machine learning using sentiment lexicons, so the part of the related works that are related to using traditional machine learning is specifically interested. As summarized in Table 1, the traditional machine learning algorithms that appears as the best performing model are random forest, logistic regression and multi-layer perceptron. Since random forest, logistic regression and multi-layer perceptron appear as the best performing algorithm for different works, so in this project, all of the three algorithms are explored. Besides these three algorithms, multinomial naïve Bayes is also explored in this project because it is known for it is designed for discrete features like BOW [7]. However, in the first experiment that uses random forest algorithm and BOW feature representation, the performance is significantly lower than the other algorithms, so random forest is dropped. The details of the experiment are described in the evaluation chapter.

Since VADER is the only sentiment lexicon used for sentiment classification task in the related works reviewed, so there is no way to compare whether VADER is performing better than other lexicons, such as TextBlob and AFINN. However, due to VADER is showing a somehow convincing results (as mention above), VADER is used in this project as the sentiment lexicon.

For feature representation, as shown in Table 1, the related works have used BOW, word embedding, TF, TF-IDF and the union of TF and TF-IDF for machine learning models. For the sentiment lexicon, raw text is used. Since there is no way to compare which feature representation is better based on the literature alone, so BOW and TF-IDF are selected based on the fact that these are common feature representation that are used in the related works.

For the number of classes of sentiment, 3 is the common number of classes used. Literature [2] proves that 3 classes is better than 5 classes, and literature [5] proves that 2 classes is even better than 3 classes. So, in this project, a two-point scale (2 classes) will be used. According to literature [5], the rating of 1-4 and the value of 5-10 from ten-point scale can be converted to negative sentiment and positive sentiment in a two-point scale respectively. In this project, this configuration is followed.

This project specifically explores adding VADER sentiment score (positive rate, neutral rate and negative rate) as extra features on top of either BOW or TF-IDF. This is not studied in the works reviewed.

IV. Proposed Solution

As mentioned earlier, for the medicine manufacturing company to track the effectiveness and side effect of new medicine after being released to the market, they can analyse the patient medication review. This proposed solution in this project focuses on the very first step before analysing the patient medication review text, which is to build a sentiment classifier to classify the sentiment of the review. Normally, positive reviews are related to effectiveness, whereas negative reviews are related to side effects. Figure 1 depicts the general process of the proposed solution.

As depicted in Figure 1, in the proposed solution, firstly every instance of the patient medication review texts is pre-processed. After pre-processing, the text is converted to five feature representations, namely VADER lexicon, BOW, BOW + VADER, TF-IDF and TF-IDF + VADER. The BOW and TF-IDF feature representation are fed into machine learning models separately. The VADER lexicon is directly mapped to sentiment class by mapping compound score above 0 as positive sentiment and otherwise as negative sentiment. BOW + VADER and TF-IDF + VADER take the combination of BOW or TF-IDF features and features from VADER sentiment lexicon (positive rate, neutral rate and negative rate). The compound score is not used because multinomial naïve Bayes cannot take negative inputs (the compound score is between -1 and 1). For each feature representation, the same supervised learning algorithms are used (multinomial naïve Bayes, logistic regression and multi-layer perceptron). All machine learning models and the VADER are evaluated on the same test set at the evaluation stage.

Figure 2 depicts the steps taken during the pre-processing stage.

As depicted in Figure 2, as the first step of pre-processing, the rows with missing value are removed. In this step, the number of instances is reduced from 280012 to 280011. After that, the rows where the rating attribute is "No rating available" are removed. In this step, the number of instances is further reduced from 280011 to 259182.

Then, the sentiment attribute is derived from the rating attribute. Ratings of 1, 2, 3 and 4 are mapped to sentiment of "negative", whereas ratings of 5, 6, 7, 8, 9 and 10 are mapped to sentiment of "positive".

As shown in Figure 2, the next step after this is cleaning the medication review text using regular expression. Table 2 lists the regular expression cleaning tasks performed in this step.

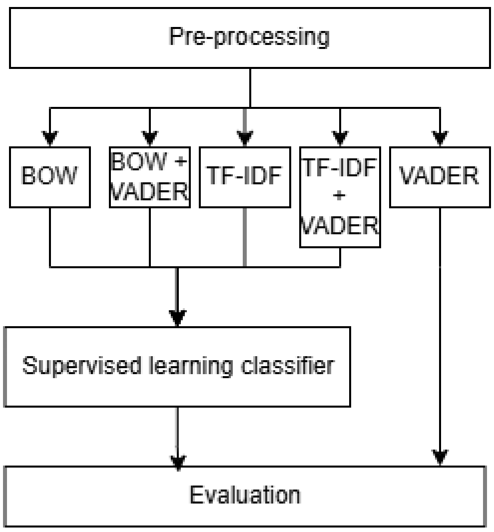

Figure 1 General process of proposed solution

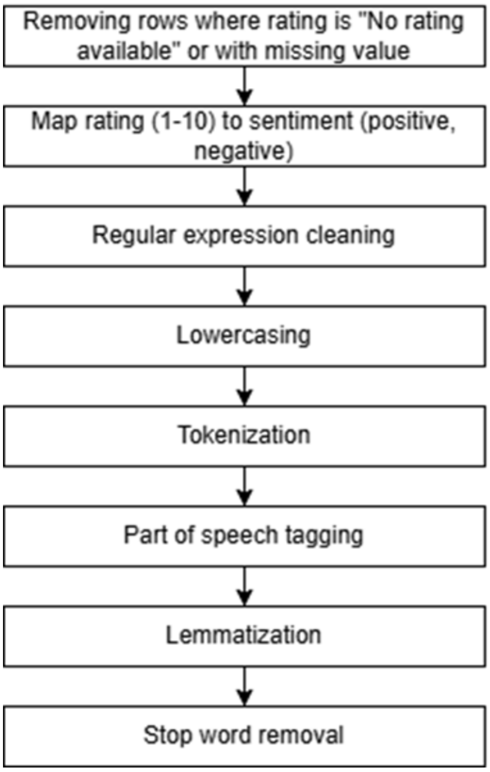

Figure 2 Pre-processing steps

Table 2 Regular expression cleaning tasks

Purpose	**Example (before)**	**Example (after)**
Change “/” to “or”	I dislike medicine A / medicine B.	I dislike medicine A or medicine B.
Change “-“ to “to”	I dislike medicine A - medicine Z	I dislike medicine A to medicine Z.
Change new line (“\n”) to white space	This medicine is good. It cures my back pain.	This medicine is good. It cures my back pain.
Change “yrs” to “years”	I take this medicine for 1 yrs.	I take this medicine for 1 years.
Change “hrs” to “hours”	I take this medicine every 4 hrs.	I take this medicine every 4 hours.
Change “x” to “times”	This medicine is 2x faster than the other medicine.	This medicine is 2 times faster than the other medicine.
Change “&” to “and”	I take this medicine & that medicine.	I take this medicine and that medicine.
Change “â€™” to “’”	Iâ€™ve taken this medicine.	I’ve taken this medicine.
Change “’ve” to “have”	I’ve taken that medicine for half years.	I have taken that medicine for half years.
Change “n’t” to “not”	I couldn’t sleep after taken this medicine.	I could not sleep after taken this medicine.
Change “I’m” to “I am”	I’m prescribed with this medicine.	I am prescribed with this medicine.
Change “’s” to “is”	It’s terrible to take this medicine.	It is terrible to take this medicine.
Change “’ll” to “will”	I’ll recommend this medicine.	I will recommend this medicine.
Change “’d” to “would”	I’d recommend this medicine.	I would recommend this medicine.
Remove punctuations (?!,()”.)	Wow! This medicine with 5mg (20%) is (super) "terrible", isnt it? 2 times more terrible.	Wow This medicine with 5mg 20% is super terrible isnt it 2 times more terrible
Remove dosage (e.g. 5mg), percentage and numbers	Wow! This medicine with 5mg (20%) is (super) "terrible", isnt it? 2 times more terrible.	Wow! This medicine with () is (super) "terrible", isnt it? times more terrible.

As shown in Table 2, in the regular expression cleaning step, the clitics are expanded to full words, the short forms are expanded to the full forms and the punctuations, dosage, percentage and numbers are removed.

The next step is lowercasing. All upper-case alphabets are changed to lower-case alphabets to reduce the number of unique words.

After that, the words are tokenized using the word_tokenize() function in Natural Language Toolkit (NLTK) library. Tokenization breaks down long paragraph into a list of single words [8].

After that, part of speech tagging is performed on the tokenized text using pos_tag() function in NLTK library. Part of speech tagging gives a syntactic overview of the sentence [8]. Part of speech tagging prepares the text for the next step, which is lemmatization because the grammar type of the word (e.g. verb, noun, etc.) is important for accurate lemmatization.

In the lemmatization step, words are reduced to its root form to reduce the number of unique words [9]. For example, the word “taking” and “taken” are reduced to “take”. In this project, lemmatization is performed using the WordNetLemmatizer() function in NLTK library.

The final step in pre-processing is stop word removal. The purpose of stop word removal is to remove the words that is semantically not significant, such as “in”, “and” and “at” [10]. This reduces the dimension (number of unique words) of the text feature.

After pre-processing, the pre-processed text is fed into VADER sentiment lexicon. VADER is a rule-based approach that refers to a pre-built lexicon which has the sentiment values for many words and uses certain rule to assign the polarity score of a sentence [11]. VADER is useful for short text, such as social media comments and user review [11]. So, VADER suits this project because patient medication review is an example of short text. In this project, SentimentIntensityAnalyzer() function from vaderSentiment library is used. Four scores, namely compound, neg, neu, and pos are generated based on each patient medication review. Compound is the polarity score normalized between -1 and 1, whereas neg is the percentage of negative words, whereas neu is the percentage of neutral words, whereas pos is the percentage of positive words [11].

Besides VADER, the pre-processed text is also converted to BOW and TF-IDF. BOW treats each word as a feature in the feature vector and the value in the vector is the number of occurrences of the word in the document [12]. In this project, CountVectorizer() from sklearn library is used. TF-IDF also treats each word as a feature, but TF-IDF assigns higher score for words that are rare across the corpus [12]. In this project, TfidfVectorizer() from sklearn library is used.

In this project, five types of feature representations are used, namely VADER compound score, BOW, BOW + VADER pos, neu and neg scores, TF-IDF and TF-IDF + VADER pos, neu and neg scores.

The VADER compound score feature is used directly to classify the sentiment of the patient medication feedback text. If the VADER compound score is greater than 0, then the instance is classified as positive, otherwise the instance is classified as negative. There is no machine learning involved in this process.

On the other hand, for the four remaining feature representations, supervised learning is used to classify the sentiment of the patient medication feedback text. Four machine learning algorithms are used, namely multinomial naïve Bayes, logistic regression, random forest and multi-layer perceptron. After the models are trained using BOW feature representation, it is found that the performance of random forest classifier is poor, with macro average F1 of 0.4075. So, for the remaining feature representations, random forest algorithm is dropped.

It is important to note that this dataset contains large number of instances, so to reduce the computation time during hyperparameter tuning, a stratified subset of the training set is used for hyperparameter tuning. The full training set is used for training after the set of best hyperparameters are found.

After all models are trained with the best parameters, to test the generalization of the models, the models are compared using the same test set. The comparison is based on the test macro average F1.

V. EVALUATION

Table 3 shows the test macro average F1 of every model developed in this project, as well as the hyperparameters used.

As shown in Table 3, the combination of multi-layer perceptron algorithm with BOW + VADER lexicon feature representation achieved the highest performance among all models, with macro average F1 of 0.8300. Note that random forest has achieved very poor macro average F1 of 0.4075 in the first experiment (BOW feature representation), so random forest was dropped for the following experiments. The possible reason is the high dimensionality of BOW features degrades the performance of random forest.

Table 3 Test macro average F1 of models (denotes the highest test macro average F1)*

Feature set	**Algorithm**	**Hyperparameters**	**Test macro average F1**
VADER compound score (baseline)	-	-	0.5884
BOW	Multinomial naïve Bayes	alpha: 0.1	0.7690
	Logistic regression	penalty: l1, C: 0.1, class_weight: balanced, solver: lbfgs, max_iter: 200	0.7950
	Random forest	n_estimators: 10, criterion: gini, max_depth: 10, min_sample_leaf: 50, min_impurity_decrease: 0.01	0.4075
	Multi-layer perceptron	hidden_layer_sizes: (10, 5), activation: logistic, max_iter: 50	0.8230
BOW + VADER pos, neu and neg scores	Multinomial naïve Bayes	alpha: 0.1	0.7695
	Logistic regression	penalty: l2, C: 0.1, class_weight: balanced, solver: saga, max_iter: 800	0.7920
	Multi-layer perceptron	hidden_layer_sizes: (15, 5), activation: logistic, max_iter: 50	0.8300 *
TF-IDF	Multinomial naïve Bayes	alpha: 0.01	0.7296
	Logistic regression	penalty: l2, C: 1.0, class_weight: balanced, solver: lbfgs, max_iter: 200	0.7970
	Multi-layer perceptron	hidden_layer_sizes: (20, 10, 5), activation: logistic, max_iter: 50	0.8035
TF-IDF + VADER pos, neu	Multinomial naïve Bayes	alpha: 0.01	0.7342
	Logistic regression	penalty: l2, C: 1.0, class_weight:	0.7968

and neg scores		balanced, solver: sag, max_iter: 100	
	Multi-layer perceptron	Hidden_layer_sizes: (10, 5), activation: logistic, max_iter: 50	0.8031

VI. Discussions

Based on the results shown in Table 3, except random forest, all machine learning models has significantly higher performance than the baseline VADER lexicon model. The performance of those models surpassed the baseline for at least 0.1412 in terms of macro average F1. This shows that machine learning is a better approach than lexicon in sentiment classification task. This is because machine learning learns the hidden pattern in the text that is related to the sentiment, whereas the lexicon is only based on the polarity of each text and does not learn any insights on the hidden pattern in the text that might be related to the sentiment.

Based on Table 3, among all machine learning algorithms, multi-layer perceptron outperformed other algorithms consistently for all feature representations. The difference in macro average F1 between multi-layer perceptron and the second well-performed model is between 0.0063 and 0.0380. This is because multi-layer perceptron is able to model complex non-linear data and often learns the hidden pattern well when the dataset is large.

Based on Table 3, it can be observed that for logistic regression, the performance on TF-IDF and TF-IDF + VADER feature representation is better than BOW and BOW + VADER feature representation. This is because TF-IDF is normalized, so it favours logistic regression which the high magnitude of inputs can degrade the performance slightly. On the other hand, for multinomial naïve Bayes and multi-layer perceptron, the performance of TF-IDF and TF-IDF + VADER feature representation is observed to be worse than BOW and BOW + VADER feature representation. This is especially significant in multinomial naïve Bayes. This is because multinomial naïve Bayes is known to be designed for raw count, so the performance degrades when the input is normalized. For multi-layer perceptron, the normalized value down-weighted the information on word frequency, though word frequency might be useful for multi-layer perceptron to extract the hidden patterns.

Based on Table 3, it can be observed that the performance can be improve or degrade slightly after adding VADER lexicon as extra feature on top of the BOW and TF-IDF feature representation. For multinomial naïve Bayes, the performance improved slightly, whereas for logistic regression, the performance degraded slightly, whereas for multi-layer perceptron, the performance improved slightly for BOW + VADER but degraded slightly for TF-IDF + VADER. The reason for improvement in performance is because VADER lexicon adds extra lexicon information of the text which BOW and TF-IDF alone cannot provide. This helps the model to learn better about the sentiment. On the other hand, the reason for degradation in performance is because VADER lexicon adds noise to the feature space. This causes the model to learn less effectively. So, it can be said that adding lexicon feature on top of BOW and TF-IDF features can sometimes improve the performance of the model slightly but sometimes introduces some noise to the feature space.

VII. Conclusion

In this project, several machine learning algorithms, namely multinomial naïve Bayes, logistic regression, random forest (dropped because of poor performance) and multi-layer perceptron, are used to build classifier to predict the sentiment of patient medication feedback text into either positive or negative. All models except random forest have surpassed the baseline VADER lexicon model in performing the sentiment classification task. The best performing model, the multi-layer perceptron with BOW + VADER lexicon feature representation, has achieved a convincing macro average F1 of 0.8300. This shows that this model is good at differentiating positive and negative sentiments in patient medication feedback text. This is very useful for the downstream task, which is to identify the effectiveness of the medicine from positive feedbacks and identify the side effects of the medicine from negative feedbacks.

Besides that, this project also investigates the improvement in the performance of the model after adding VADER lexicon information to the conventional feature representation of BOW or TF-IDF. The experiment results showed that the intuition that additional lexicon information can enhance the performance of the classifiers is only partially correct, because additional lexicon information may also add additional noise to the conventional BOW or TF-IDF feature representation. That is why the change on the performance brought by adding VADER lexicon information is not consistent. It does improve the performance for some cases - in fact, the best performing model comes from BOW + VADER lexicon feature representation. However, it is not guaranteed that adding lexicon information will improve the performance, as there are 3 out of 6 cases that the performance has degraded.

References

[1] M. Kumar and T. Singh, "Patient Insights: 2.8Lakh Drug & Condition Reviews [Data set]," Kaggle, 2024. [Online]. Available: https://doi.org/10.34740/KAGGLE/DSV/9196455. [Accessed 3 December 2025].

[2] C. Allenbrand, "Supervised and unsupervised learning models for pharmaceutical drug rating and classification using consumer generated reviews," HealthcareAnalytics, vol. 5, 2024.

[3] A. B. Nair, A. K., A. U., D. T. Jaison, A. V. and V. S. Anoop, ""Hey..! this medicine made me sick": Sentiment analysis of user-generated drug reviews using machine learning techniques," 9 April 2024. [Online]. Available: https://doi.org/10.48550/arXiv.2404.13057. [Accessed 3 December 2025].

[4] W. E. Bottacin, A. Luquetta, L. Gomes Jr, T. T. d. Souza, W. C. T. Reis and A. C. Melchiors, " Sentiment analysis in medication adherence: using ruled based and artificial intelligence driven algorithms to understand patient medication experiences," International Journal of Clinical Pharmacy, vol. 47, pp. 970-980, 2025.

[5] A. S. Azar, S. B. Rikan, A. Naemi, J. B. Mohasefi and U. K. Wiil, "Predicting patients' sentiments about medications using artificial intelligence techniques," Scientific Reports, vol. 14, 2024.

[6] E. Saad, S. Din, R. Jamil, F. Rustam, A. Mehmood, I. Ashraf and G. S. Choi, "Determining the Efficiency of Drugs UnderSpecial Conditions From Users Reviews on Healthcare Web Forums," IEEE Access, vol. 9, 2021.

[7] Scikit Learn, "MultinomialNB," Scikit Learn Documentation, [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.naive_bayes.MultinomialNB.html. [Accessed 7 December 2025].

[8] S. Arora, "NLTK - NLP," Geeks for Geeks, 8 August 2025. [Online]. Available: https://www.geeksforgeeks.org/python/nltk-nlp/. [Accessed 7 December 2025].

[9] Y. S. Rautela, "Lemmatization with NLTK," Geeks for Geeks, 4 October 2025. [Online]. Available:

https://www.geeksforgeeks.org/python/python-lemmatization-with-nltk/. [Accessed 7 December 2025].

[10] kartik, "Removing stop words with NLTK in Python," Geeks for Geeks, 7 October 2025. [Online]. Available: https://www.geeksforgeeks.org/nlp/removing-stop-words-nltk-python/. [Accessed 7 December 2025].

[11] AnkThon, "Sentiment Analysis using VADER - Using Python," Geeks for Geeks, 11 July 2025. [Online]. Available: https://www.geeksforgeeks.org/python/python-sentiment-analysis-using-vader/. [Accessed 8 December 2025].

[12] S. Sharma, "Bag-of-words vs TF-IDF," Geeks for Geeks, 23 July 2025. [Online]. Available: https://www.geeksforgeeks.org/nlp/bag-of-words-vs-tf-idf/. [Accessed 9 December 2025].